**Politecnico di Milano**

# Dipartimento di Elettronica, Informazione e Bioingegneria

**prof. Fabrizio Ferrandi**

## Advanced Algorithms and Parallel Programming 1st part–September 10th, 2026

**Polimi ID** _____ _____

**Surname** _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

**Exercise 1 (4 points)** _____

**Exercise 2 (4 points)** _____

**Exercise 3 (4 points)** _____

**Exercise 4 (4 points)** _____

**Exercise n. 1**

A. What does it imply that each memory cell can hold an integer of unbounded size in a PRAM? (1)

B. In the PRAM model, explain the meaning of work efficiency and why a parallel algorithm may perform more work than its sequential counterpart. Give one concrete example involving parallel overhead. (1)

C. Please discuss what type of PRAM memory accesses are performed in the vector sum PRAM algorithm. (1)

D. Please outline the key aspects of weak scalability (1)

**Exercise n. 2**

A. For randomized Quicksort, explain how the expected running time is obtained from the recurrence induced by a random pivot. Discuss why the worst case remains quadratic. (2)

B. Describe one Divide and Conquer example discussed during the lectures and explain why the Master Theorem is useful for analyzing this class of algorithms. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

C. What is memoization, and why is it useful in Dynamic Programming? Give an example of a repeated subproblem for which memoization avoids recomputation. (1)

**Exercise n. 3**

A. Why can randomization be useful in a data structure even when the required answer must always be correct? Give one example. (1)

B. What does it mean for a sorting algorithm to be stable, and why is stability important in Radix Sort? (1)

C. Randomization can simplify balancing and give good expected performance while preserving correctness. True/False? Example? (2)

**Exercise n. 4**

A. Describe the ideas behind the primality test algorithm presented during the lectures. Which class of randomized approach does it belong to? (2)

B. Describe the potential method of amortized analysis and apply it conceptually to a dynamic array that doubles when full. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**